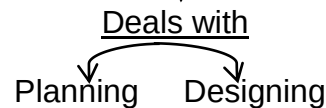
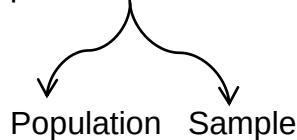
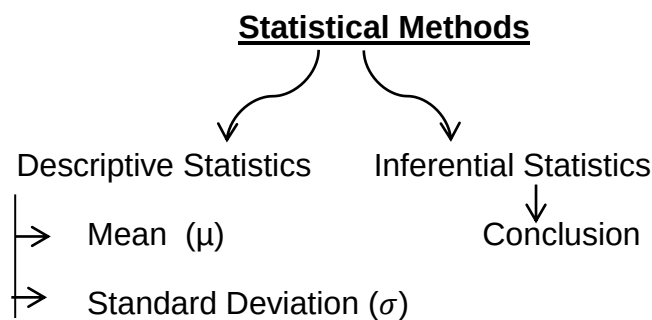


Statistics

➡ Statistics is the branch of mathematics where we collect, organize, analyse and represent the data for better decision making. We apply statistics to different problems.



Data – Facts or pieces of information that can be measured.



Descriptive Statistics :

Descriptive Statistics is a summary that describes or summarizes the collection of information/data.

It summarizes the sample data rather than learning about the population that sample data is representing.

Inferential Statistics :

Inferential Statistics is the process of data analysis where we make the conclusions about population data using sample data.

Population (N) :

Population Data is the entire group that you want to draw conclusions about.

Sample (n) :

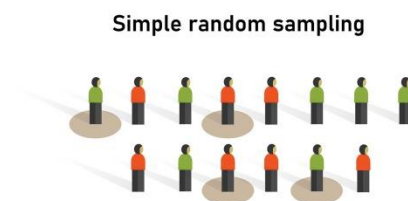
Sample is the group (part of population) from which you'll collect data.



➡ Types of sampling :

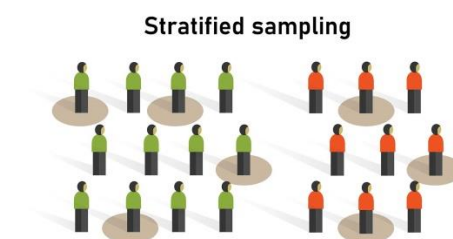
1. Simple Random Sampling

Simple Random Sampling is the process of sampling where every member of the population has equal chance of being selected.



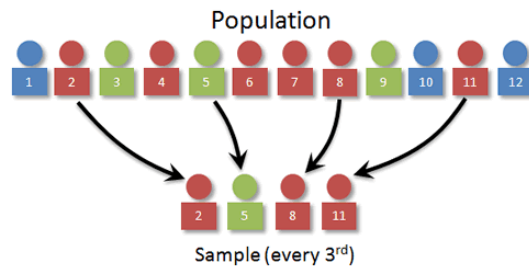
2. Stratified Sampling

Stratified sampling is a method of sampling where population (N) is split into non-overlapping group.



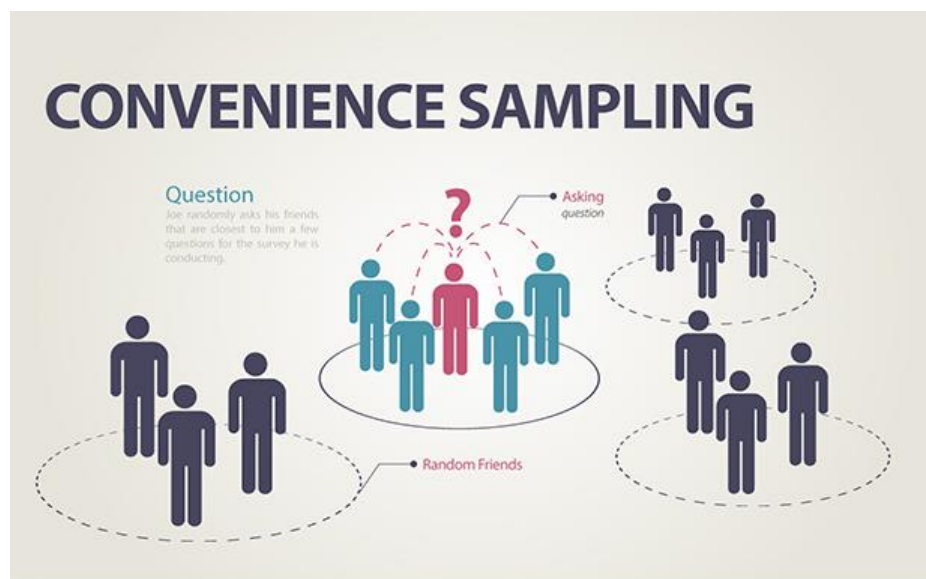
3. Systematic Sampling

Systematic sampling is a probability sampling method where researchers select members from population at nth interval.



4. Convenience Sampling

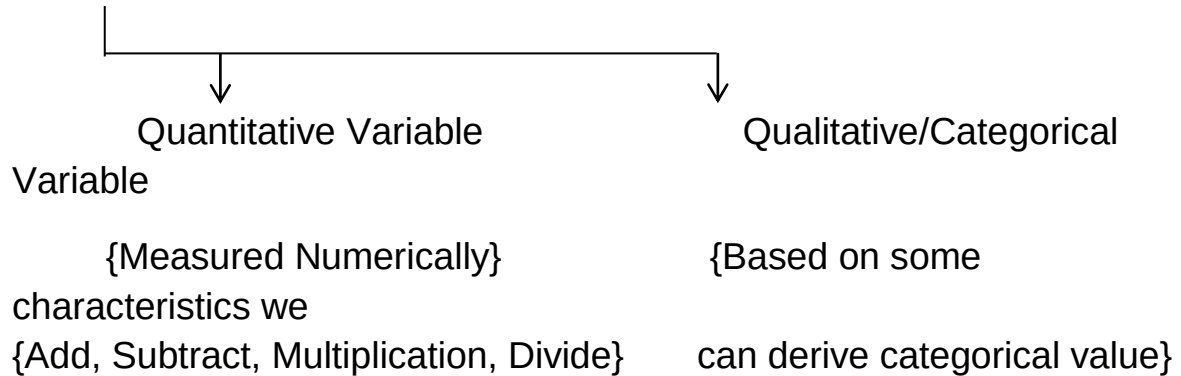
Convenience sampling is the process of taking sample data from those who has knowledge/expertise on the research area.





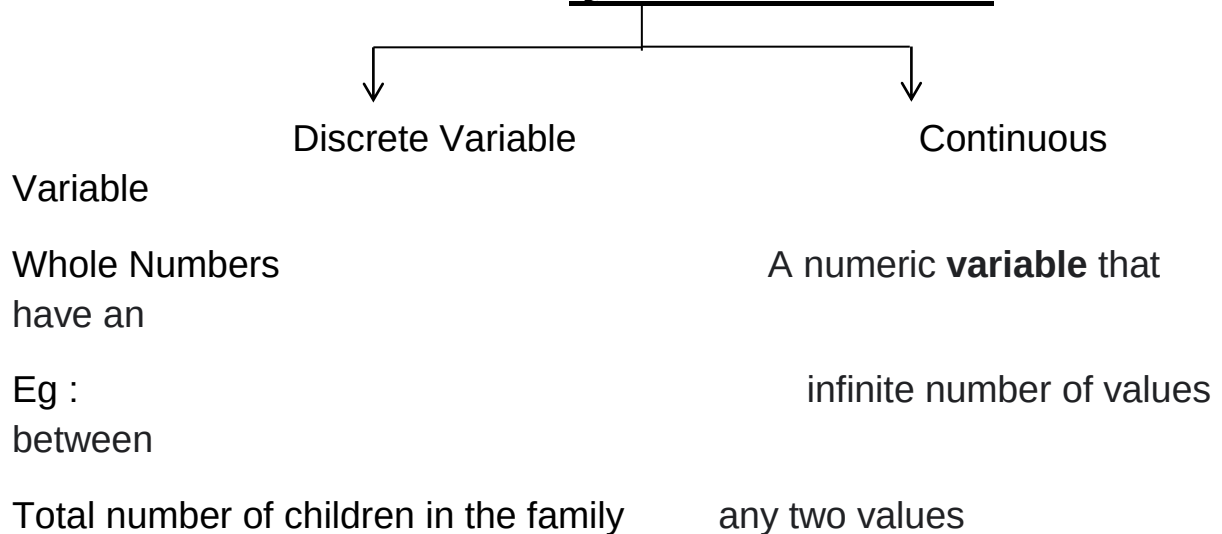
Variables

A variable is a property that can take any value.



Quantitative Variables	Qualitative Variables
<i>Take on numeric values</i>	<i>Take on names or labels</i>
Number of students in a class	Eye color
Number of square feet in a house	Gender
Population size of a city	Breed of dog
Age of an individual	Level of Education
Height of an individual	Marital status

Quantitative Variable



2, 4, 5

Eg :

Height – 172.5, 163.9,

162.8

➡ Variable Measurement Scale

There are 4 types of measured variables

1. **Nominal** : Categorical data
2. **Ordinal** : Order of the data matters but value doesn't.
3. **Interval** : Order matters, value also matters but natural zero is not present.
4. **Ratio** : Something measured on a ratio scale, has same properties as interval scale but with absolute zero point.

Eg :

<u>Students (Marks)</u>	<u>Rank</u>
100	1
96	2
57	3
85	4
44	5

← Ordinal Data

Eg :

Temperatures

Fahrenheit

70 – 80

80 – 90

90 – 100

0

Interval
Zero doesn't make any useful meaning

Ratio Variable :

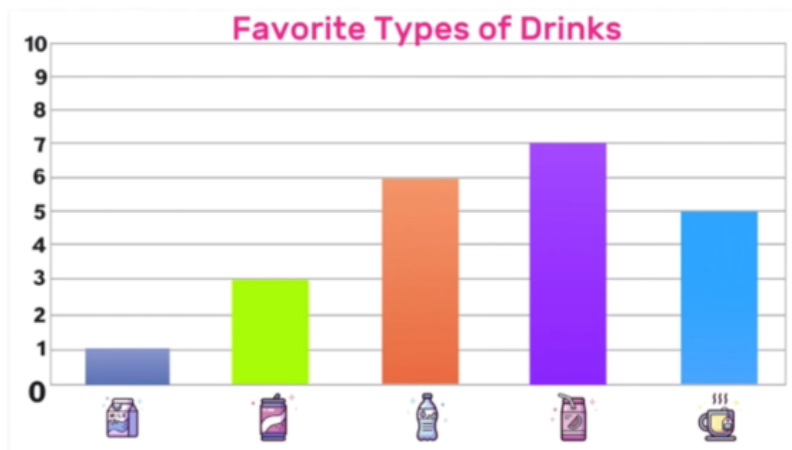
Provides more detailed informations. It includes true zero value.

➡ **Frequency Distribution :**

Sample Dataset : Rose, Lily, Sunflower, Rose, Lily, Sunflower, Rose, Lily, Lily

<u>Flower</u>	<u>Frequency</u>	<u>Cumulative Frequency</u>
Rose	3	3
Lily	4	7
Sunflower	2	9

➡ **Bar-graph :** (Categorical)

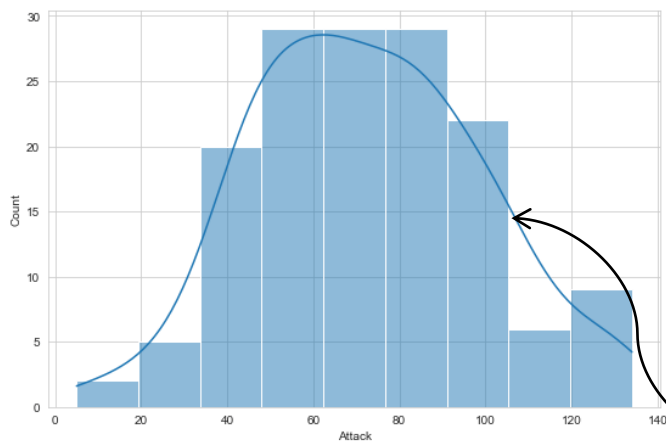


➡ **Histogram :** (Continuous)

It is a graphical tool to summarize discrete or continuous data.

Eg:

Ages : {10, 12, 14, 18, 29, 26, 30, 35, 36, 37, 40, 41, 42, 43, 50, 51}



Histogram

To construct histogram :

Bins/Bucket ←

If data set starts from zero.

Otherwise = $(\text{max} - \text{min}) / \text{bins}$

pdf : Smoothing of the histogram.

{Probability density function}

{Kernel Density Estimator}

➡ Measure of central Tendency :

1. Mean
2. Median
3. Mode

Data

Quantitative

Refers to the measure used to determine the centre of the distribution of the dataset.

{1,1,2,2,3,3,4,5,5,6,100}

Mean = $(32+100)/11 = 12$

Median

Sort the numbers

Odd = n Middle element

even = $(n1 + n2)/2$ Middle two element

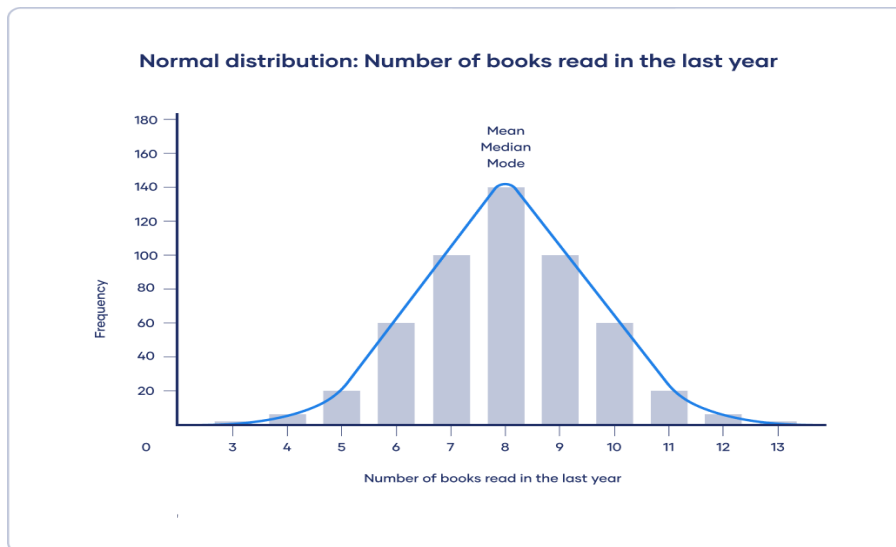
Median works well with outliers.

Mode

Most frequent element.

Eg :

{1,2,2,3,4,5,6,6,6,7,8,100,200,100} Mode = 6

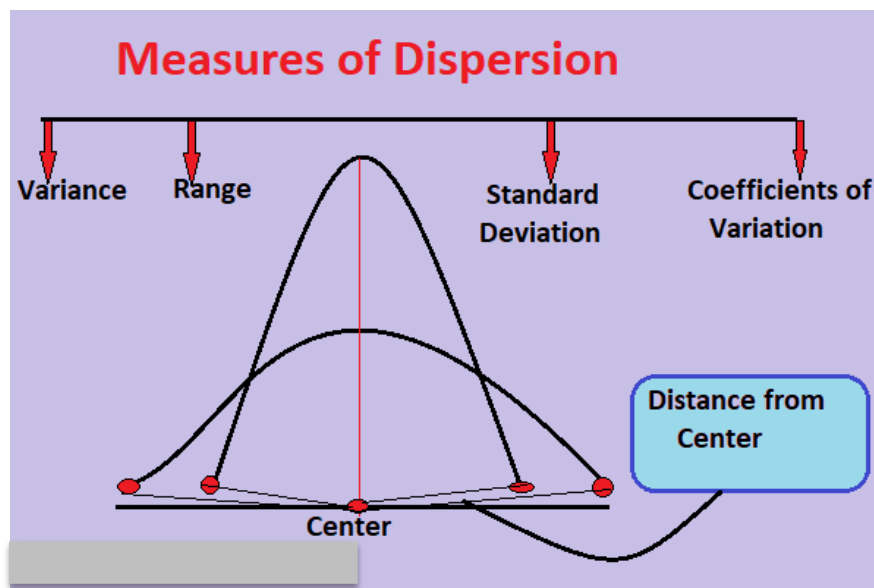


➔ Measure of Dispersion :

Measure of dispersion describes the spread of the data or its variation around the centre value.

1. Variance
2. Standard Deviation

{Dispersion}
↓
Spread



1. Variance :

Population Variance :

$$\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$

Sample Variance :

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

Spread is low means the elements present in the central region is more.

More variance : Data is more spread.

Variance = Spread = Dispersion = Is the extent to which distribution is stretched or squeezed.

2. Standard Deviation :

$$\sigma = \sqrt{\text{variance}}$$

Standard deviation shows how far the elements are from mean.

➡ Percentiles and Quartiles : {find outliers}

Percentile is a value below which a certain percentage of observation lie.

Eg :

Dataset : {2,2,3,4,5,5,5,6,7,8,8,8,8,8,9,9,10,11,11,12}

What is the percentile range of 10 ?

n = 20

$$\begin{aligned} \text{Percentile rank of } x &= \frac{(\# \text{ of values below } x)}{x} * 100 \\ &= \frac{16}{20} * 100 = 80\% \end{aligned}$$

What value exists at percentile ranking of 25% ?

$$\begin{aligned}\text{Value} &= \frac{\text{Percentile}}{100} (n + 1) \\ &= \frac{25}{100} (21) = 5.25 \text{ index} \\ &\approx \frac{5+5}{2} = 5\end{aligned}$$

➡ **5 Number Summary**

1. Minimum
2. First Quartile
3. Median
4. Third Quartile
5. Maximum

Removing Outliers

{1,2,2,2,3,3,4,5,5,5,6,6,6,6,7,8,8,9,27}

$$Q1 = \frac{25}{100} (20) = 5\text{th index} = 3$$

$$Q3 = \frac{75}{100} (20) = 15 \text{ index} = 8$$

$$\text{Lower fence} = Q1 - 1.5(\text{IQR}) = -4.5$$

$$\text{Higher fence} = Q3 + 1.5(\text{IQR}) = 15.5$$

$$\text{Inter Quartile Range (IQR)} = Q3 - Q1 = 8 - 3 = 5$$

Remaining data : {1,2,2,2,3,3,4,5,5,5,6,6,6,6,7,8,8,9}

Minimum : 1

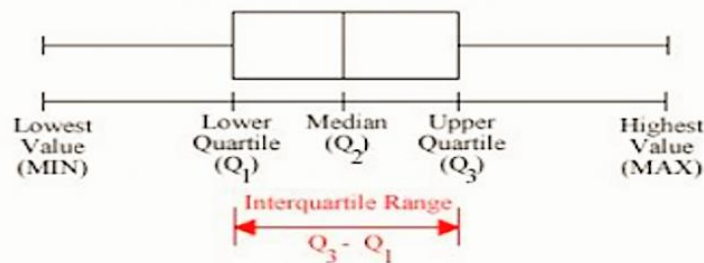
First Quartile : 3

Median : 5

Third Quartile : 8

Maximum : 9

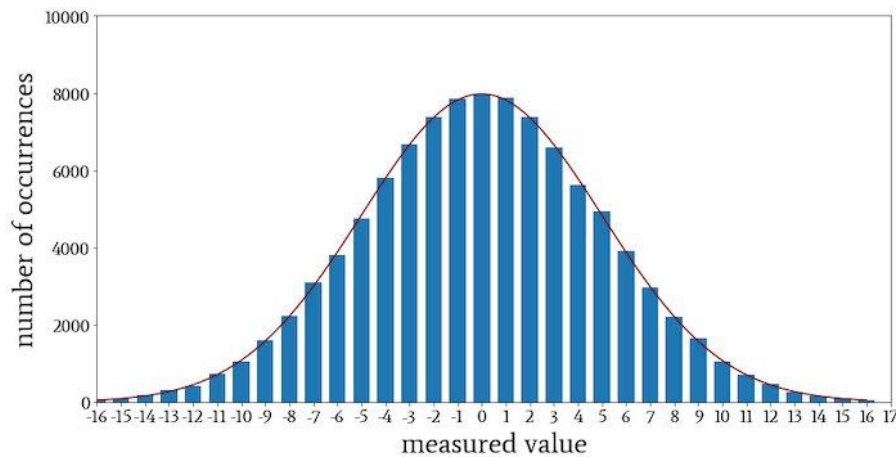
➡ Boxplot



➡ Normal Distribution :

A distribution is called normal distribution if we plot histogram with the data it'll be symmetric to the mean and dataset are more frequent towards the mean.

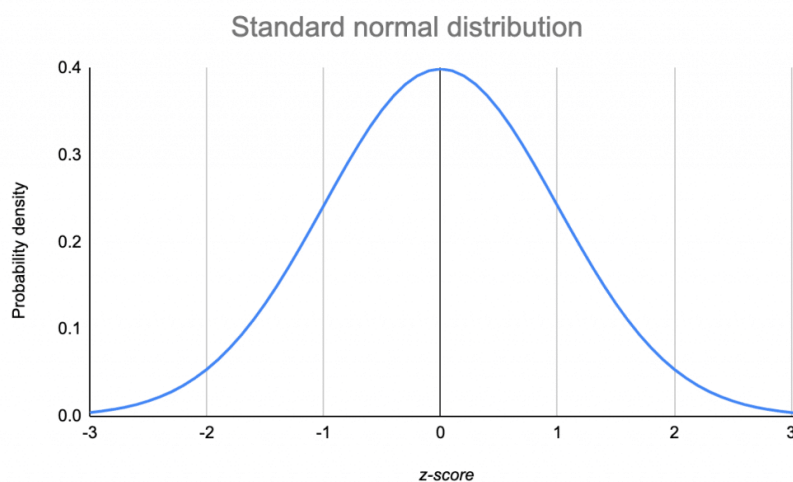
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



➡ Standard Normal Distribution :

A distribution is called standard normal distribution where mean will be zero and standard deviation will be one. And most of the data will lie in between -3σ to $+3\sigma$.

If the dataset will be in different unit ML algorithm will take more time. So we will apply SND on these dataset to bring them to the same scale.



➡ Standardization :

In standardization we scale down the value, where the mean will be zero and standard deviation will be one.

➡ Normalization :

In normalization we try to convert a dataset in between a range.

1. MinMax Scaler :

With the help of MinMax scaler we convert the data which will range in between 0 to 1.

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

➡ Covariance :

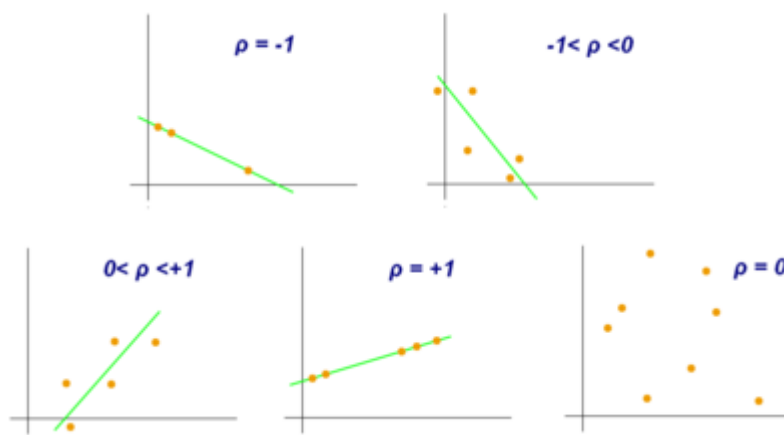
Covariance helps us to find-out the direction of relationship.

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_j - \bar{Y})}{n}$$

➡ Pearson Correlation Coefficient :

Pearson Correlation coefficient measures the strength and relationship between two variables. The value ranges in between -1 to +1. The values more towards +1 they are more positively correlated and the values more towards -1 the more negatively correlated they are.

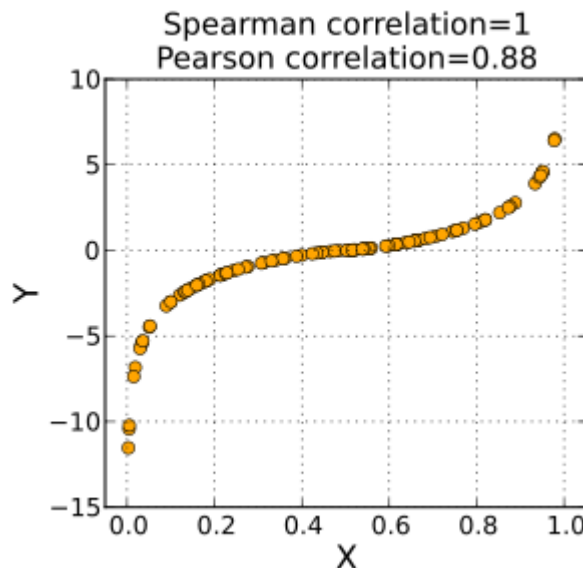
$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$



➡ Spearman's rank correlation coefficient :

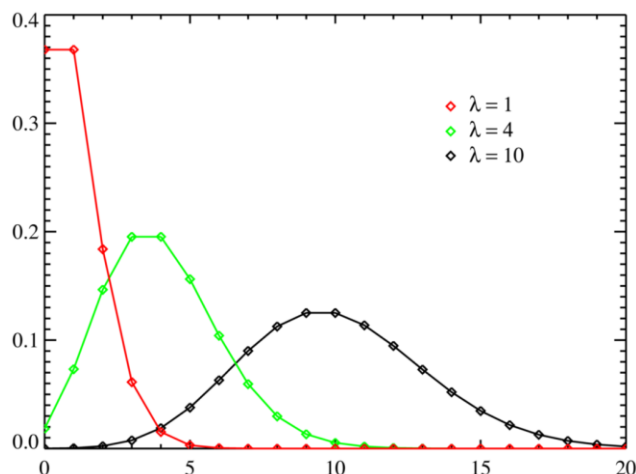
Spearman's correlation assesses monotonic relationships (whether linear or not). If there are no repeated data values, a perfect Spearman correlation of +1 or -1 occurs when each of the variables is a perfect monotone function of the other.

$$r_s = \rho(r_x, r_y) = \frac{\text{covariance}(r_x, r_y)}{\sigma_{r_x} * \sigma_{r_y}}$$
$$= \rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$



➡ Poisson Distribution :

Poisson distribution is a probability distribution which gives the probability of an event can occur in a given interval of time.



3 conditions of Poisson distribution :

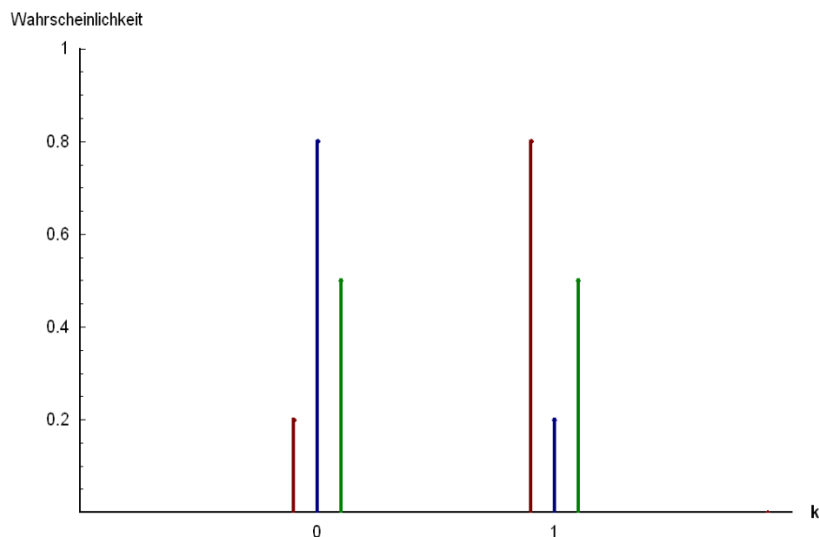
- **Events are independent of each other.** The occurrence of one event does not affect the probability of another event will occur.
 - The average rate (events per time period) is constant.
 - Two events cannot occur at the same time.
-

$$f(x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

➡ Bernoulli Distribution :

Bernoulli distribution is a discrete probability distribution which takes the value 1 with probability p and the value 0 with probability q = 1-p. It is a set of possible outcomes of any single experiment.

$$f(x) = \begin{cases} p^x * (1-p)^{1-x} & \text{if } x = 0,1 \\ 0 & \text{otherwise} \end{cases} = \begin{cases} p & \text{if } x = 1 \\ 1-p & \text{if } x = 0 \end{cases}$$

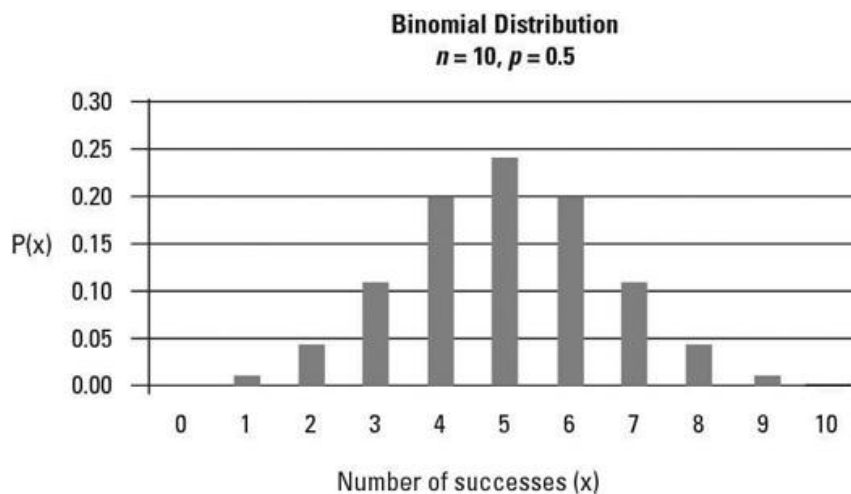


➡ Binomial Distribution :

Binomial Distribution is a discrete probability distribution which represents number of success in n independent experiments.

Binomial Distribution Formula

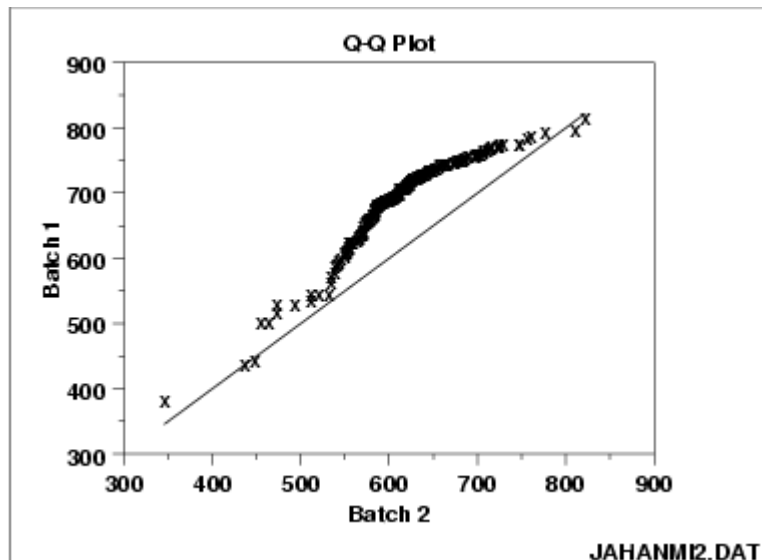
$$P(X) = {}_n C_x p^x (1 - p)^{n-x}$$



➡ Q-Q plot :

Q-Q plot is a probability plot for compare and analyze two probability distributions by plotting their quantiles against each other. If the two distributions which we are comparing are exactly equal then points on Q-Q plot perfectly lie in straight line $y=x$.

With the help of Q-Q plot we can identify a distribution is Gaussian or not.



→ **Chi-square test :**

Chi-square test claims about population proportion.

It is a non-parametric test that is performed on categorical variable.

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$$\chi^2 = \sum \frac{(f_o - f_E)^2}{f_E}$$

f_o = observed frequencies
 f_E = expected frequencies

→ **Hypothesis Testing :**

Hypothesis testing is a method used to decide whether the data to draw the inference about the parameter is true or not.

z-test

We go for t-test if

i. We know the population std deviation. or

ii. We don't know the population std deviation but our sample size is greater than 30 ($n > 30$).

$$z\text{-statistic} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

t-test

We go for t-test if

- i. We don't know the population std deviation.
- ii. Our sample size is less than 30 ($n < 30$).
- iii. Sample std deviation is given.

$$t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}}$$

➡ Annova Test :

An ANNOVA (analysis of variance) test is a statistical test used to determine the statistical difference between two or more categorical groups by calculating mean by using variance.

$$SS_{total} = \sum_{j=1}^p \sum_{i=1}^{n_j} (x_{ij} - \bar{x})^2$$
$$SS_{between} = \sum_{j=1}^p n_j (\bar{x}_j - \bar{x})^2$$
$$SS_{within} = \sum_{j=1}^p \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_j)^2$$

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There are different types of ANOVA tests

- i. One way
 - ii. Two way
-

The difference between these two types depends on the number of independent variables in the dataset.

A one-way ANOVA has one categorical independent variable and a normally distributed continuous variable.

A two-way ANOVA has two or more categorical independent variables and a normally distributed continuous dependent variable.

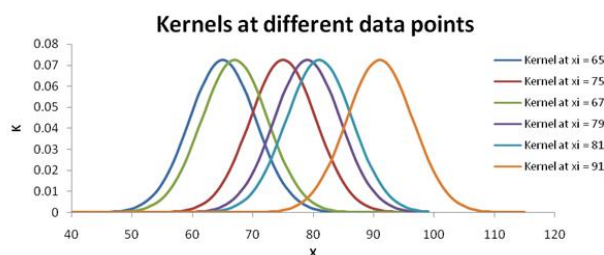
➡ 1 sample t-test and 2 sample t-test :

A **one-sample t-test** is used to compare a single population to a standard value (for example, to determine whether the average lifespan of a specific town is different from the country average).

A **paired t-test** is used to compare a single population before and after some experimental intervention or at two different points in time (for example, measuring student performance on a test before and after being taught the material).

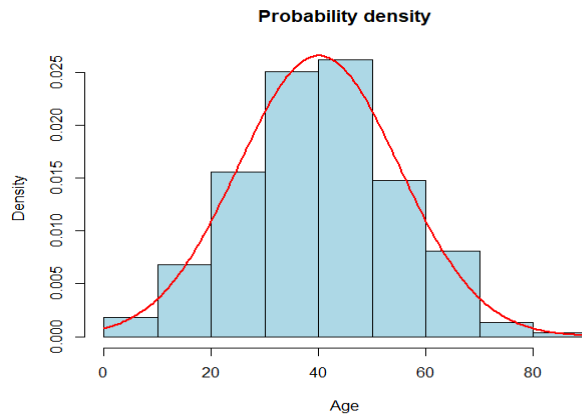
➡ Kernel density estimation :

Kernel density estimation (KDE) is a smoothing process where inferences about population are made based on finite sample value and main aim of KDE is to estimate probability density function for the given dataset.



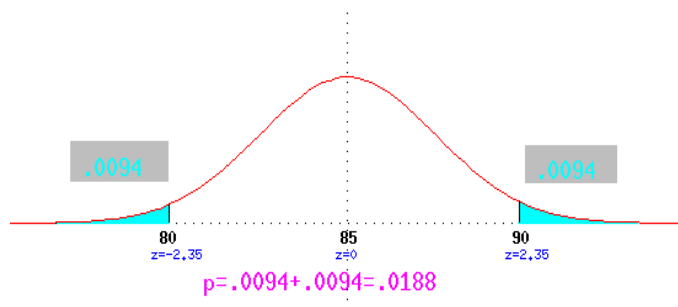
➡ Probability density function :

Probability density function (PDF) is nothing but a probability function which represents the density of continuous random variable lying in between a range of values.



➔ **P-value :**

P-value always represents the significance level. It tells you how many values are not contributing out of whole experiments. (in general words p-value tells you how many experiments are going to fail out of 100)



➔ **Bell curve :**

A bell curve is a graph which describes the normal distribution and has a shape similar to a bell.

The top of the curve shows mean, median and mode of the dataset.

